Methods | MTH1002 | Term 2
Dr. Mike O'Sullivan | Year: 2025-2026

Exercise Sheet 2 (Tutorials in Week 4)

Key:

- **Questions marked with a ☕** are for discussion in your tutorials.
- **Questions marked with a 🦇** are difficult questions.
- **Questions marked with a ♫** have been marked so that they don't feel left out. Please don't leave them out during your independent learning and revision time.

☕ **Exercise 1.** Find f_x, f_y, f_{xy}, f_{xx} and f_{yy} for the functions

(i) $f(x,y) = x^2y + x^4y^3$;

(ii) $f(x,y) = \sin(x+y^2)$;

(iii) $f(x,y) = xy^2 - x\ln(xy)$.

☕ **Exercise 2.** Verify that $y(t,x) = e^{-kn^2t}\sin nx$ satisfies

$$y_t = ky_{xx};$$

☕ **Exercise 3.** Find the gradient of the following functions at the given point and calculate the directional derivative in the direction of the unit vector $\mathbf{u}$.

(i) $f(x,y) = y + x^2/y$ at (1,2), $\mathbf{u} = \frac{12}{13}\mathbf{i} + \frac{5}{13}\mathbf{j}$.

(ii) $g(x,y) = x^2 - 3xy + 2y^2$ at (1,1), $\mathbf{u} = \frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}$.

(iii) $h(x,y) = x\tan^{-1}(y/x)$ at (1,−1), $\mathbf{u} = \frac{2}{\sqrt{5}}\mathbf{i} + \frac{1}{\sqrt{5}}\mathbf{j}$.

♫ **Exercise 4.** If $u = u(\xi,\eta)$ and $\xi = x + ay, \eta = x + by$, find the values of a and b such that they transform the partial differential equation

$$\frac{\partial^2 u}{\partial x^2} + 4\frac{\partial^2 u}{\partial x \partial y} + 3\frac{\partial^2 u}{\partial y^2} = 0$$

into

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = 0.$$

Find the general solution of the partial differential equation.

♫ **Exercise 5.** Verify that $r(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ satisfies the equation

$$r_{xx} + r_{yy} + r_{zz} = 2/r.$$

♫ **Exercise 6.** Find the value of $\partial z/\partial x$ at the point $(1, 1, 1)$ if the equation

$$2xy + z^3x - yz = 2$$

defines z as a function of the two independent variables x and y.

♫ **Exercise 7.** The temperature $T(x, y, z)$ in a room is given by $T(x, y, z) = x^2 + y^2 - z^2$.

1. Find the gradient of $T(x, y, z)$ at the point $(1, 1, -2)$.
2. Write the equation of the tangent plane to $T(x, y, z)$ at this point.